18 $16^{\frac{1}{5}} \times 2^{x} = 8^{\frac{3}{4}}$

Work out the exact value of x.

Write both 16 and 8 as powers of 2.

$16 = 2^4$ and $8 = 2^3$

1 mark
Use a common base

Multiply powers in a power; add powers in a product.

$\frac{4}{5} + x = \frac{9}{4}$

1 mark
Equate the exponents of the common base 2

-4/5 -4/5

$x = \frac{9}{4} - \frac{4}{5} = \frac{45 - 16}{20} = \frac{29}{20}$

1 mark
Exact value 29/20, equivalent to 1.45

1.45

(Total for Question 18 is 3 marks)